

FAR WESTERN UNIVERSITY
Semester End Examination-2080

Engineering Mathematics II (Integral + Differential Calculus) SH121

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. Discuss the meaning of consistency and inconsistency of system of linear equation $AX = B$ with example. Test the consistency of system of equation, $3x + 6y + z = 16$, $2x + 4y + 3z = 13$, $x + 3y + 2z = 9$ and solve if possible by back substitution method. (2+4+2)
2. Find the distance from the point (3, 4, 5) to the point where the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$, meets the plane $x + y + z = 2$.
3. Define the parametric equation with example and discuss why parametric equation are assumed to be more efficient way than Cartesian equation to represent a path. A particle $p(x, y)$ moves in a plane with $x = a \cos t$, $y = -b \sin t$, $0 \leq t \leq \pi$, discuss the motion and path traced out by the particle. (1+2+5)
4. Discuss about the necessity of reducing double integral in Cartesian co-ordinate to polar co-ordinate, with example. Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{xdydx}{\sqrt{x^2+y^2}}$, by reducing into polar co-ordinate. (2+6)
5. Discuss the meaning of solution of differential equation with example and also discuss the process of formation of differential equation corresponding to family of curve, $f(x, y, c_1, c_2) = 0$. Form the differential equation corresponding to family of curve, $xy = A e^x + B e^{-x}$. (2+2+4)
6. Define linear differential equation of first order and first degree with example. Solve the differential equation, $(x^3 + 1)\frac{dy}{dx} + 3x^2y = \sin^2x$. (2+6)

Group – 'B'

Attempt any four questions.

(Long answer questions)

4×15=60

7. Define linear transformation and inverse linear transformation with example. Show that a linear mapping $T: V \rightarrow W$ is one to one iff $\text{Ker } T = \{0\}$. Also discuss the meaning of invertible linear transformation and mention the criteria for linear transformation to be invertible, show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (3x-y, 4x+2y)$ is invertible. Also find formula for T^{-1} . (2+4+7+2)
8. Define three dimensional locus, cone and cylinder and mention the criteria that they reduces to right cone and right cylinder. Deduce the equation of cone having vertex at (α, β, γ) and guiding curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, z = 0$. Also find equation of right cone having vertex at origin and axis $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$, with vertical angle 30° . (2+1+7+5)
9. Define regular and irregular singular point of the differential equation $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$. Find the regular and irregular singular points of $x^2(x-2)^2 \frac{d^2y}{dx^2} + 2(x-2) \frac{dy}{dx} + (x+3)y = 0$. Also find the series solution of $x(x-1) \frac{d^2y}{dx^2} + (3x-1) \frac{dy}{dx} + y = 0$, about $x = 0$. (2+3+10)
10. Define the n^{th} order differential equation with constant coefficient. Also describe the process of finding solution of differential equation $\frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = Q(x)$. Show that if $y = f_1(x)$ and $y = f_2(x)$ are two independent solutions of $\frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$, then $y = A f_1(x) + B f_2(x)$ is also solution of same equation. Find the solution of $(D^2 - 4D + 4)Y = x^2 + e^{2x}$. (2+4+9)
11. (a) Discuss the geometrical meaning of double integral $\iint_C f \, dA$. Evaluate the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 \, dx \, dy$, by changing the order of integration. (8)
- (b) Define sphere and find the equation of sphere passing through points $(3, 0, 2), (-1, 1, 1), (2, -5, 4)$ and having Centre on the plane $2x + 3y + 4z = 6$. (7)

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Object Oriented Programming (CT123)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. Explain how the object oriented programming exhibits encapsulation? How objects oriented programming is different than procedure oriented programming? (3+5)
2. Briefly explain the terms namespace and enumerator. In what condition the inline functions are preferred over normal functions? Write a suitable program to make inline function of a class. (3+2+3)
3. Write some characteristics of static data. Write a suitable program to describe the operation of static function. (4+4)
4. Write a suitable program to show conversion from user defined data type to user defined data type conversion.
5. What is input output stream? Explain the input output stream class hierarchy. Differentiate between the file accessing mechanisms. (1+4+3)
6. Write a program to overload binary plus (+) operator by making overloading function as a friend function.

P.T.O.

Group – 'B'

Attempt any four questions.

(Long answer questions)

4×15=60

- ✓7. A) Explain the features of C++ and write about its history. (4+4)
B) Write syntax to allocate memory for different data types dynamically. Write a suitable program for dynamic memory allocation of any object. (2+5)
8. A) Write some characteristics of virtual functions. Write a meaningful program to show passing of an object to function. (5+5)
B) Write a program to illustrate the operation of virtual destructor. (5)
9. A) Write some rules for operator overloading. (7)
B) Define the multipath inheritance write its syntax. Explain the hierarchical inheritance and write a program that describes its operation. (2+6)
10. A) Describe how the constructors and destructors are invoked in multiple inheritances if base and derived class have parameterized constructor and write suitable program for it. (7)
B) Define inheritance and explain multilevel and hybrid inheritance. (1+7)
- ✓11. A) Explain how the templates follow generic programming? Explain and write a program to overload normal function with the template function. (3+4)
B) Define an exception. Write the advantages of exception handling mechanism over conventional error handling mechanism. Write a suitable program that explains any of the exception handling mechanism. (1+4+3)

FAR WESTERN UNIVERSITY
Semester End Examination-2080

Study Skills in English for Academic Purposes (EAP) SH125

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. National governments are good at prioritizing, so they should also decide the order in which global problems are dealt with. Do you agree with this statement? Explain.
2. Do tropical cyclones have any positive effects? To what extent it can be taken as positive?
3. Write down the advantages of pictograms and ideograms.
4. Do you think Hofstede's four dimension of culture exist in your locality? Which of the dimensions have been seen in effect and how?
5. What is plagiarism? How can you avoid unintentional plagiarism when revising your draft?
6. Why should reading be done with a purpose and a particular reason? Explain.

Group – 'B'

Attempt any four questions.

(Long answer questions)

4×15=60

7. What is cartography? How can the accuracy be ensured while preparing maps? Explain.
8. Do you think that the way that a person is raised can affect their gender identity? Cite specific cases as an example to ensure your point of view.
9. What is the metaphor glass ceiling used for? What are its effects? Discuss with some specific cases.
10. What is the purpose of note taking? Explain the different ways of recording information for the further and future use.
11. Mention the different frames used for enhancing writing skills. Explain any one of the frame in details providing an example.

FAR WESTERN UNIVERSITY
Semester End Examination-2080

Fundamental of Thermodynamics and Heat Transfer (ME127)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. Define thermodynamic property. Explain intensive and extensive property with examples. Sketch polytropic processes with $n=0$, 1, γ and ∞ on a common P-V diagram. Derive an expression for work transfer for polytropic process. [4+2+2]
2. What is quasi-static process? Differentiate between stored energy and transient energy with examples. [4+4]
3. Define pure substance. State and explain "state postulate". Derive the relation $\bar{v} = v_f + x v_{fg}$ for two phase mixture. [1+3+4]
4. A piston-cylinder device contains 0.05m^3 of a gas initially at 200kPa. At this state, a linear spring constant of 150kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas, causing the piston to rise and to compress the spring until the volume inside the cylinder doubles. If the cross-sectional area of the piston is 0.25m^2 , determine:
 - i. The final pressure inside the cylinder
 - ii. The total work done by the gas
5. A piston cylinder device shown in **Fig.1** below contains water initially at a pressure of 125kPa with a quality of 50%. Heat is added to the system until it reaches to a final temperature of 800°C . It takes a pressure of 600kPa to lift the piston from stops. Sketch the process P-V and T-V diagram and determine:
 - i. The mass of H_2O in the system and
 - ii. The total work transfer

6. The compression ratio of an air standard Otto cycle is 8. At the beginning of the compression process, the pressure and temperature of air are 100kPa and 22°C respectively. The heat added per kg of air during the cycle is 2100kJ/kg. Determine:
- The pressure and temperature at the end of each process of the cycle
 - The thermal efficiency
 - The mean effective pressure
- [Take $R=287\text{J/kgK}$ and $c_v = 718\text{J/kgK}$]

Group – 'B'

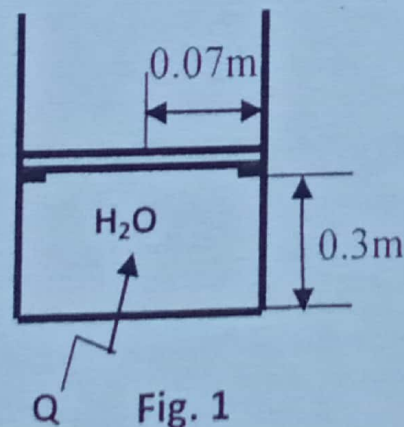
Attempt any four questions.

(Long answer questions)

4×15=60

7. State and explain conservation of energy for control volume. Steam enters through a turbine operating under steady state condition at a rate of 5kg/s. The properties of steam at turbine inlet are $P_1 = 1\text{MPa}$ and $T_1 = 500^\circ\text{C}$ and at the turbine exit $P_2 = 50\text{kPa}$ and $x_2 = 0.9$. If the heat loss from the turbine surface occurs at a rate of 500kW, determine the power output from turbine. [8+7]
8. Define heat engine, heat pump and refrigerator. Explain how first and second law is applied to determine performance of heat engine and refrigerator. A Carnot engine operates between two reservoirs at temperature T_L and T_H . The work output of the engine is 0.6 times the heat rejected. The difference in temperature between the source and the sink is 200°C. Calculate the thermal efficiency, source temperature and the sink temperature. [8+7]
9. Draw the layout diagram, P-v and T-s diagram of air standard diesel cycle and derive an expression for its efficiency in terms of compression ratio and cut-off ratio. An ideal Brayton cycle has pressure ratio of 10. The temperature of air at the compressor and turbine inlets are 300K and 1200K respectively. Determine its thermal efficiency and mass flow rate of air to produce net power out of 80MW. [Take $C_p=1005\text{J/KgK}$, $\gamma = 1.4$] [7+8]

- ✓ 10. Derive an expression with appropriate diagram for conduction heat transfer through a composite cylinder tube consisting of three layers of different materials. A hollow cylinder with inner and outer diameter of 8 cm and 12 cm respectively has an inner surface temperature of 200°C and outer surface temperature of 50°C . If the thermal conductivity of the cylinder material is 60W/mK , determine the heat transfer from the unit length of pipe. Also determine the temperature at the surface at a radial distance of 5 cm from the axis of the cylinder. [7+8]
- ✓ 11. Derive expressions for change in entropy for an ideal gas and incompressible substance. Steam enters a nozzle operating at steady state with $P_1 = 10\text{bar}$, $T_1 = 400^{\circ}\text{C}$ and velocity of 10m/s . The steam flows through the horizontal adiabatic nozzle. At the exit, $P_2 = 1.5\text{bar}$ and the velocity of 1068.13m/s . The mass flow rate is 2kg/s . Determine the exit area of the nozzle in m^2 . [Refer the table for properties of water] [8+7]



FAR WESTERN UNIVERSITY
Semester End Examination-2080
Construction Materials (CE126)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. As a civil engineer, what are the major factors which you consider most for the selection of material in civil engineering project in Nepal? Write short notes on significant mechanical properties of civil engineering materials.
2. List out and explain with neat sketch the various defects found in timber.
3. What is lime cycle? Explain manufacturing process of lime in detail.
4. Describe in short about different iron ores.
5. What is corrosion? What are the factors affecting corrosion and how it can be controlled?
6. Explain various ingredients of a good paint. List and explain various types of paints.

Group – 'B'

Attempt any four questions.

(Long answer questions)

4×15=60

7. a. With neat sketches, explain in detail about the manufacturing process of conventional bricks. [12]
b. Differentiate between lime and cement. [3]
8. a. Explain the various types of tests on building stones and provide briefly specifications for good building stone. [12]
b. Differentiate between setting and hardening of cement. [3]
9. List the fresh and hardened properties of concrete. Explain any two for each fresh and hardened properties of the concrete with neat sketch.
10. a. Write down the reaction in the kiln during manufacture of the cement. [6]
b. What is steel? What are the reasons for the manufacture of steel? How can you identify the steel and wrought iron? [4]
c. Describe various classification of glass and its properties. [5]
11. Write short notes on: [3×5=15]
 - a. Bouge's compounds
 - b. Mineral admixtures
 - c. Electrical, Thermal & Sound Insulating Materials

FAR WESTERN UNIVERSITY

Semester End Examination-2080

Physics (Mechanics and Optics) SH122

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – 'A'

Attempt any five questions.

(Short answer questions)

5×8=40

1. a) Define phase of a particle. Give the relation between phase difference and path difference. (2+2)
b) What is interference? What will happen in interference fringe in double slit experiment if one slit is covered? (1+3)
2. a) What is polarization? Differentiate between quarter wave plate and half wave plate. (1+3)
b) Write the features of LASER? Write the differences between spontaneous emission and stimulated emission. (1+3)
3. a) Define Young's modulus, modulus of rigidity and Poisson's ratio, and write relation between them. (4)
b) Explain why a hollow cylinder is stronger than solid cylinder of the same length, mass and material. (4)
4. a) What is Reynold number? Explain significance of Reynold's number. (1+3)
b) What is elasticity? Discuss elastic hysteresis. (1+3)
5. a) Derive equation of continuity for steady flow of liquid. (4)
b) What is surface tension? Explain why a liquid is either raised or depressed in the capillary tube? (1+3)
6. a) Discuss the double refraction of light. (4)
b) Differentiate between diffraction and interference of light. (4)

Group – 'B'

Attempt any four questions.

(Long answer questions)

4×15=60

7. a) What is cantilever ? Derive an expression for the depression of loaded end of a cantilever of circular cross-section of negligible weight. (2+8)
- b) A pipe is running full of water. At a certain point A it tapers from 0.6m diameter to 0.2m diameter at B. The pressure difference between A and B is 1m of water column. Find rate of flow of water through the pipe. (5)
8. a) Discuss Poiseuille's assumptions and derive Poiseuille's formula for the rate of flow of a liquid through tube. (2+8)
- b) Calculate the work done in spraying a drop of mercury of 1 mm radius into one million identical drops all of the same size. (Surface tension of mercury is $550 \times 10^{-3} \text{ N/m}$) (5)
9. a) What is zone plate ? Show that how zone plate acts as converging lens of multiple foci. (2+8)
- b) Calculate the highest order spectrum which may be seen with monochromatic light of wavelength $6000 \text{ \AA}$ by means of a diffraction grating with 5000 lines/cm. (5)
10. a) What is Fresnel's biprism? Describe necessary theory to determine wavelength of light by using Fresnel biprism. (1+9)
- b) Two slits are 0.3 mm apart and placed 50cm from screen. What is distance between the second and third dark line of the interference pattern when the slits are illuminated with a light of 600nm wavelength. (5)
11. a) Describe working principle of He-Ne gas laser. (5)
- b) Derive the formula for acceptance angle for optical fiber. (5)
- c) In an optical fiber, the core material has refractive index 1.6 and refractive index of clad material is 1.4. Calculate the values of critical angle and angle of acceptance. (5)

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Engineering Drawing II (AR124)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks. Assume suitable data if necessary.

Full Marks: 60

Time: 3hrs.

Attempt all questions.

6×10=60

1. Draw the isometric view of figure 1. [8]
2. Draw oblique view of the following figure 2. [8]
3. Draw and name the type of fit for 60 S6/h12. [Given: S=-0.053, 6=0.019, h=0.000 & 12=0.3000] [6]
4. Draw the Hexagonal Nut of standard M24. [8]
5. Draw perspective view of the following figure 3. [6]
6. Draw the detail drawing of given Centering Cone from figure 4. [10]
7. Draw the Assembly drawing of given wheel support from figure 5. [14]

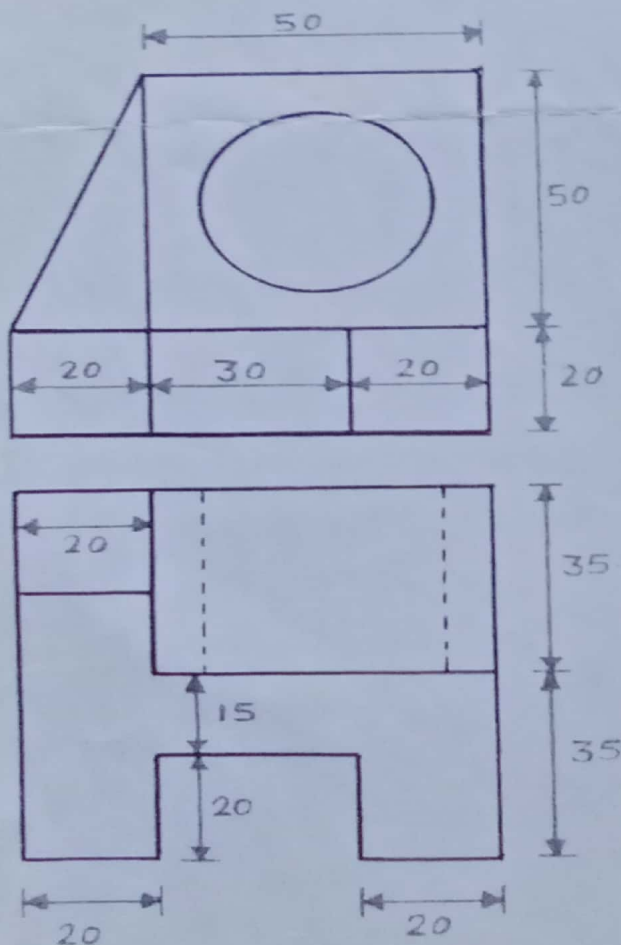


Figure 1

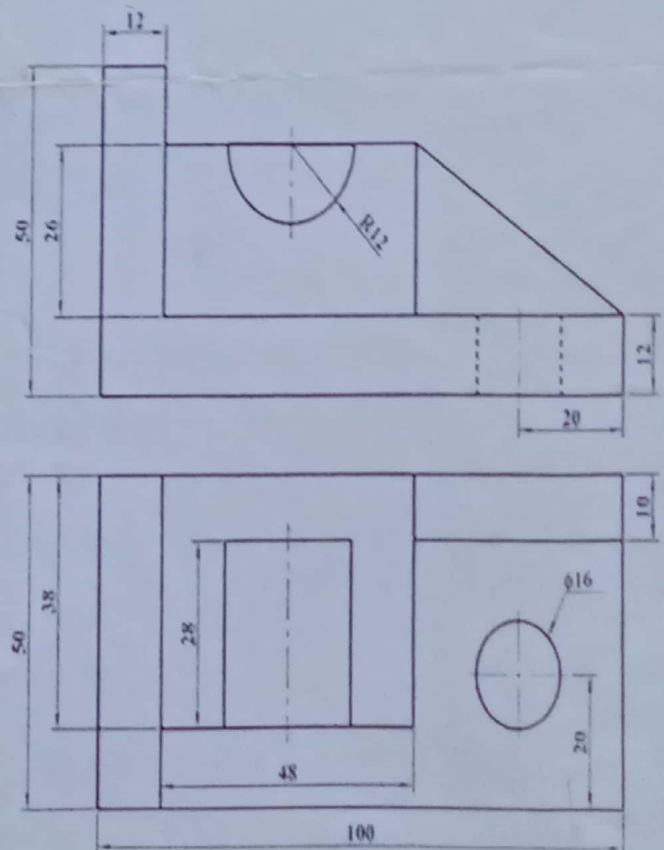


Figure 2

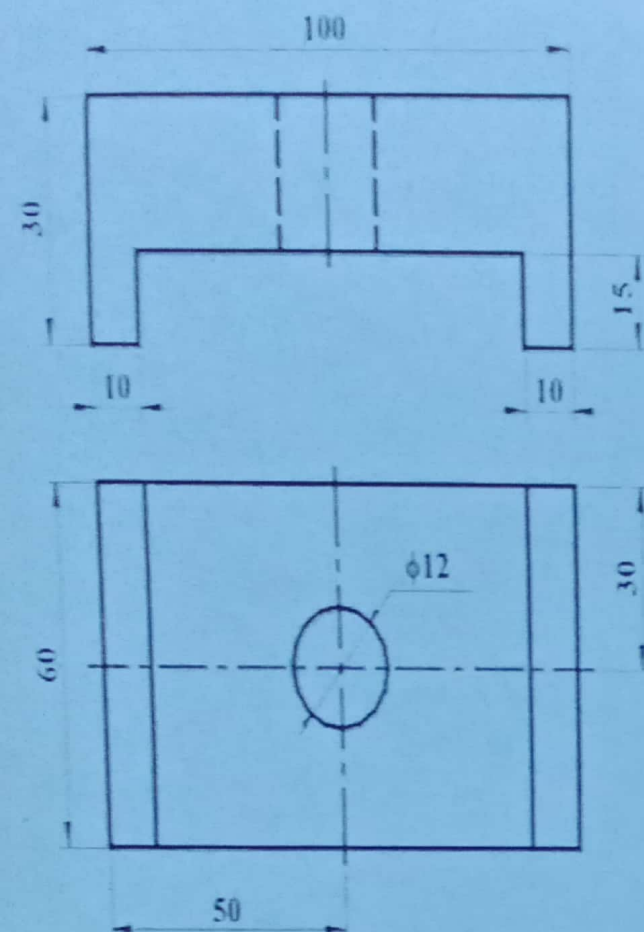


Figure 3

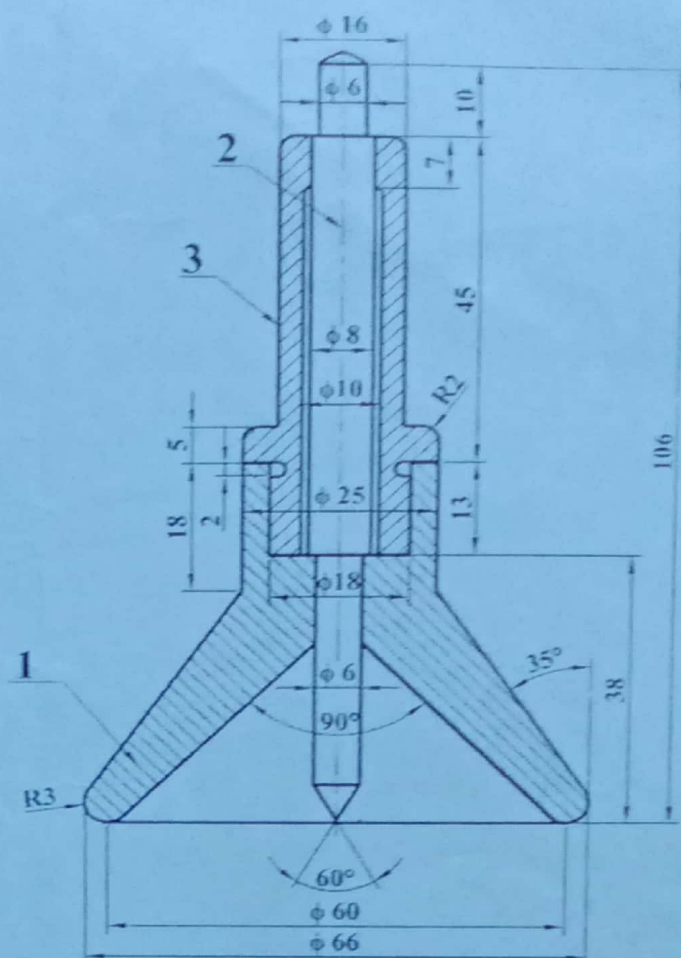


Figure 4

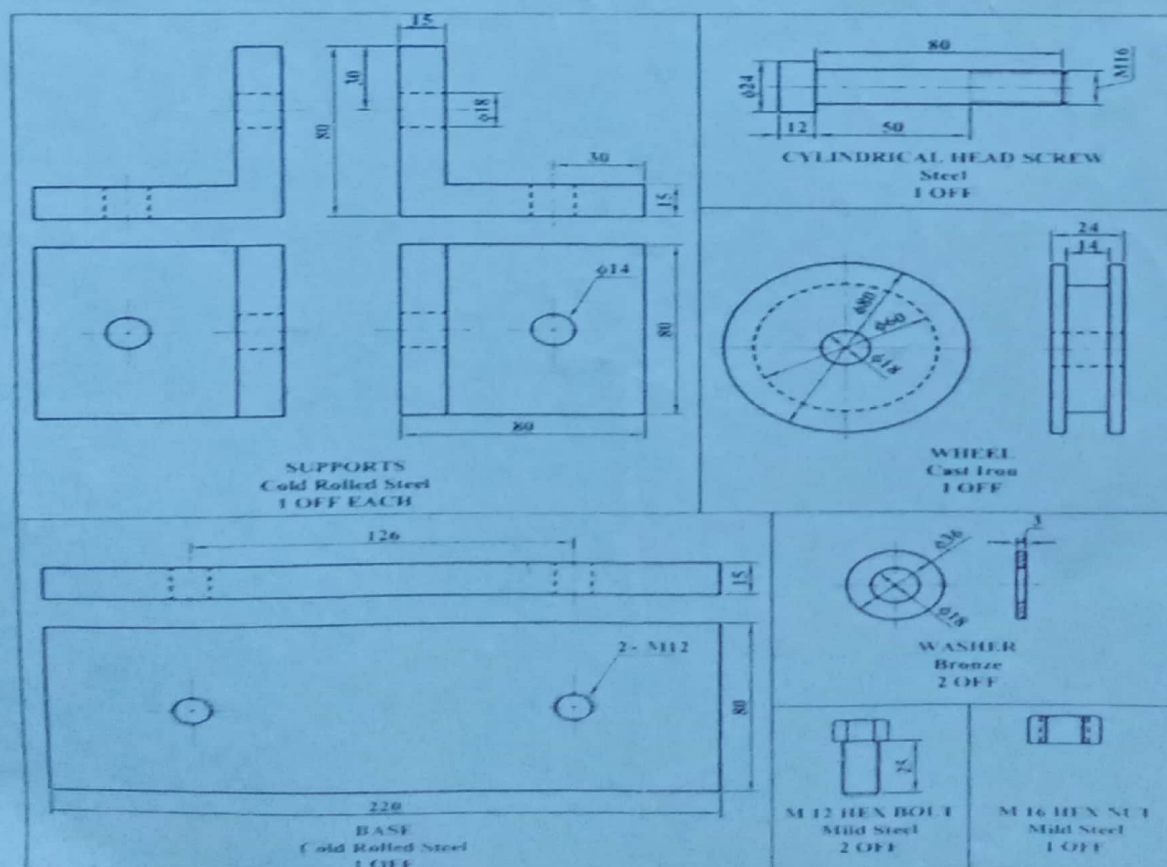


Figure 5